

Inverse functions



1 Graph $y = x$ on the same axes as the graphs of the two functions and look to see one of the graphs is a reflection of the other about $y = x$. If they are reflections, then they are inverses.

2

a No

b Yes

c No

d Yes

e No

f Yes

g No

3

a

i $f^{-1}(x) = \frac{1}{8}(x + 9)$

ii $[-41, 7]$

iii $[-4, 2]$

b

i $f^{-1}(x) = \sqrt{x}$

ii $[0, \infty)$

iii $[0, \infty)$

c

i $f^{-1}(x) = \sqrt{16 - x^2}$

ii $[0, 4]$

iii $[0, 4]$

4

a Yes

b No

c Yes

d Yes

e Yes

f Yes

g Yes

h No

5

a Yes

b No

c Yes

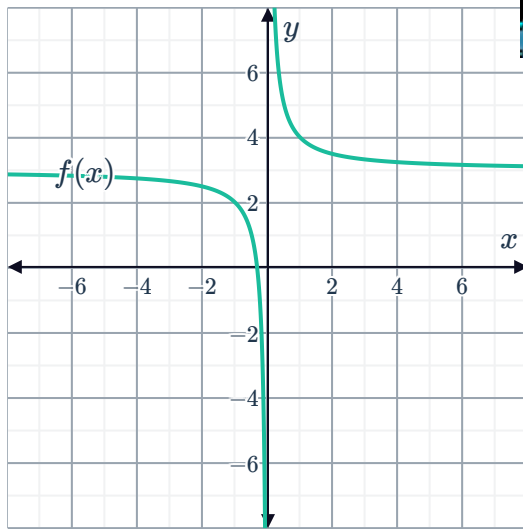
d Yes

e No

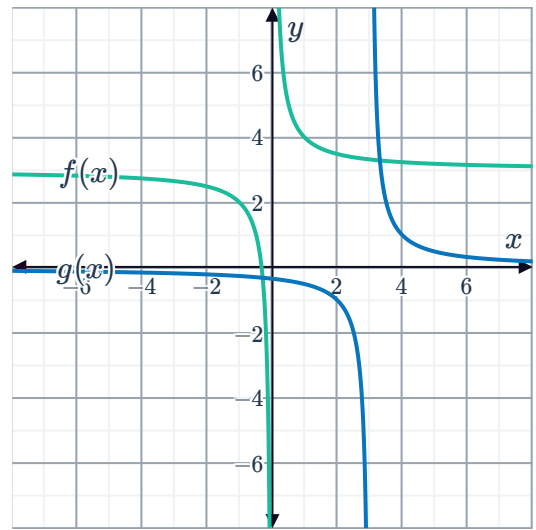
Graphs of inverse functions

6

a

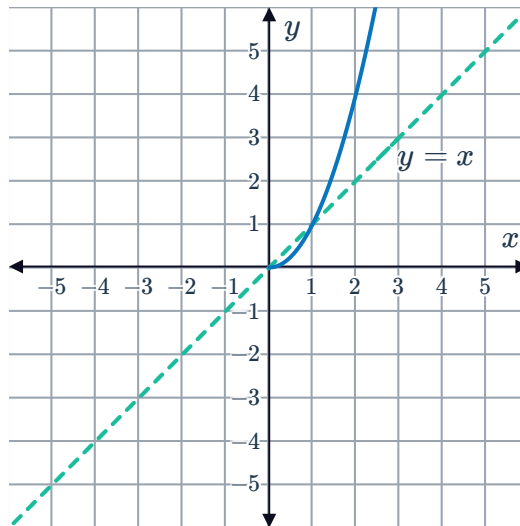


b



c Yes

7



8

a $x^3 - 2$

b $\sqrt[3]{x+2}$

c x

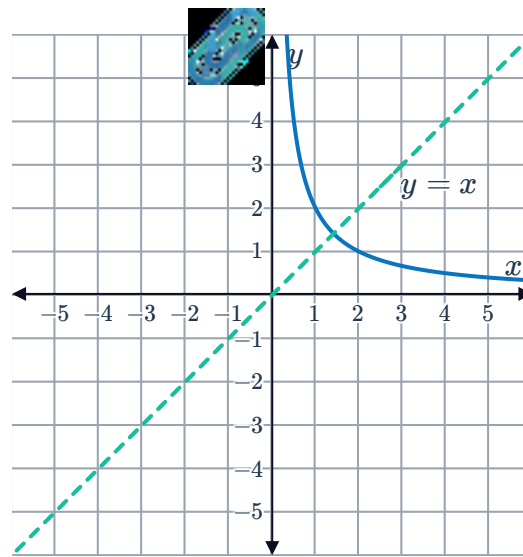
d x

e

i Yes ii No iii Yes iv Yes

9

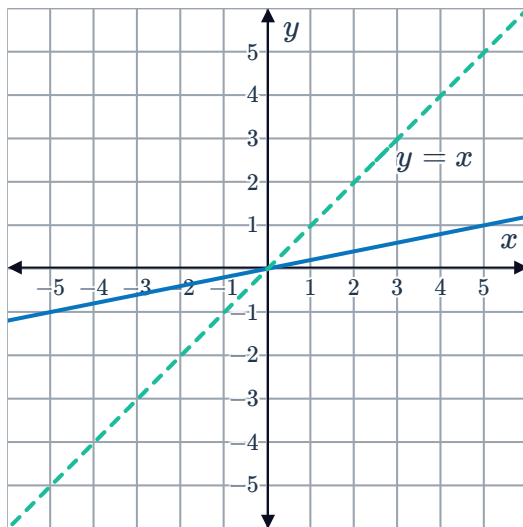
a



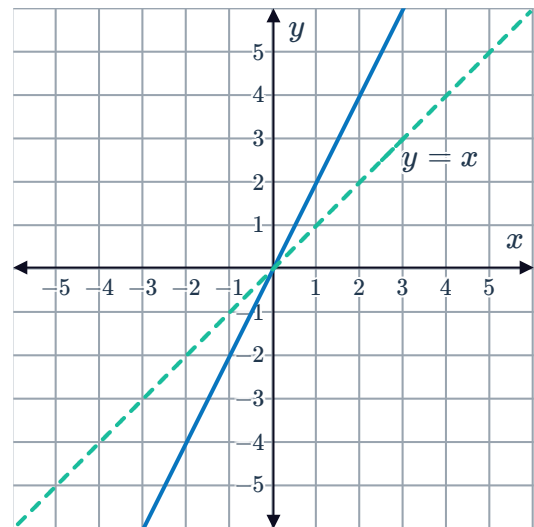
b The inverse graph is exactly the same as the original graph.

10

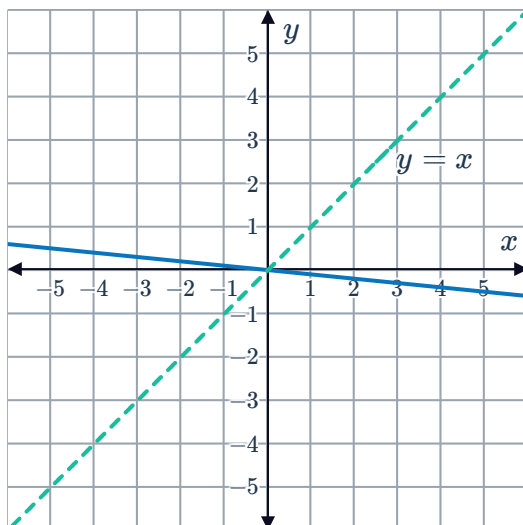
a



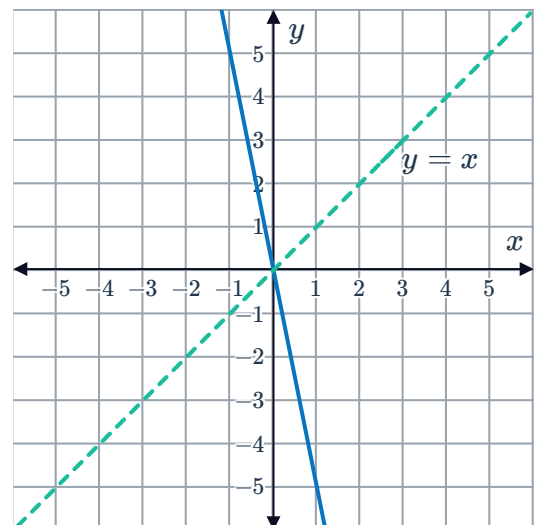
b



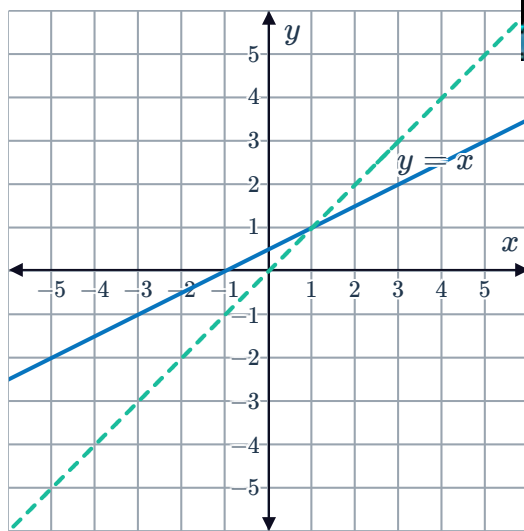
c



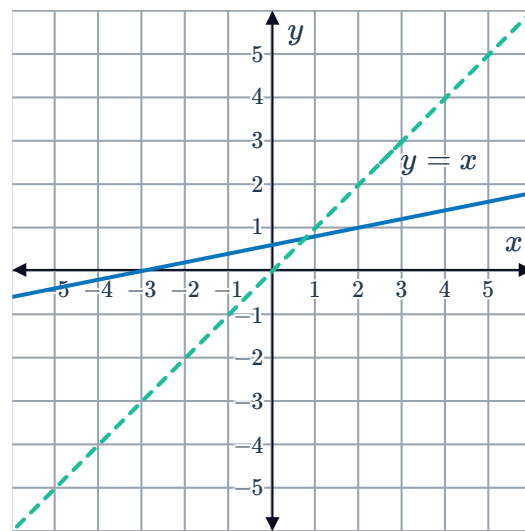
d



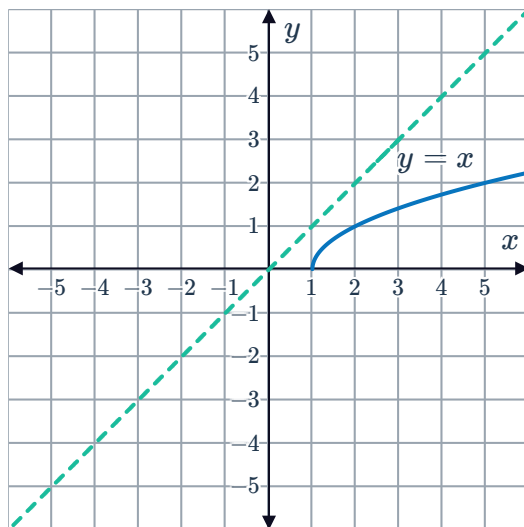
e



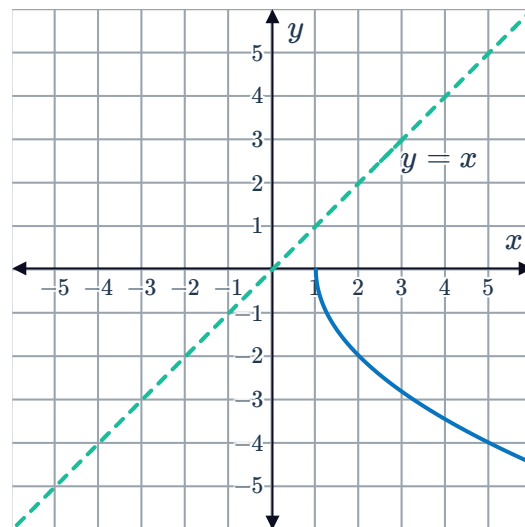
f



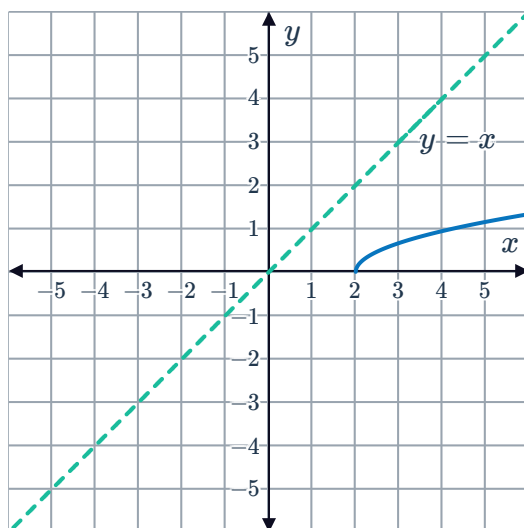
g



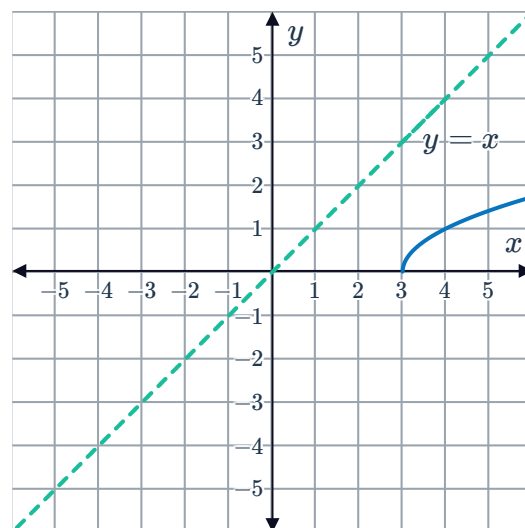
h



i

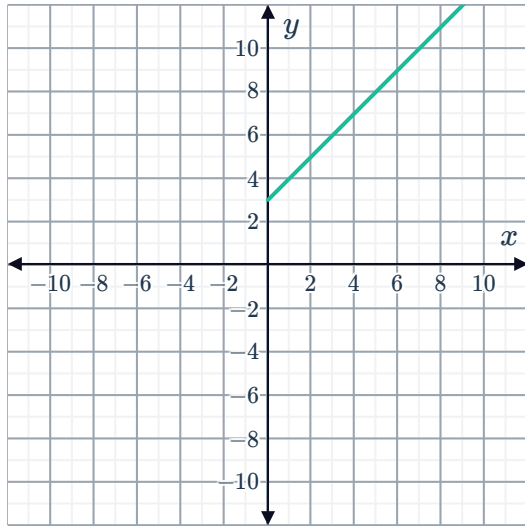


j



a

i



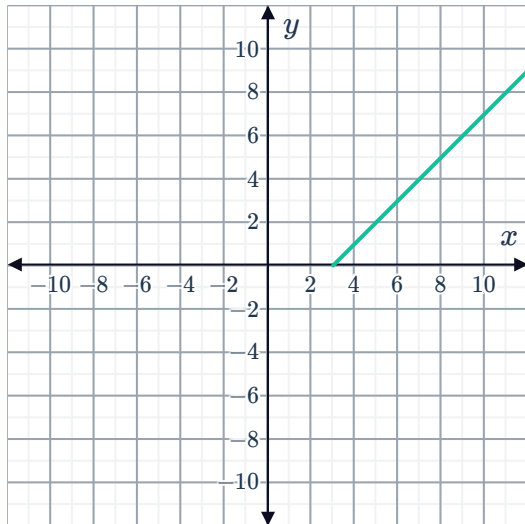
ii

$$f^{-1}(x) = x - 3$$

iii $[3, \infty)$

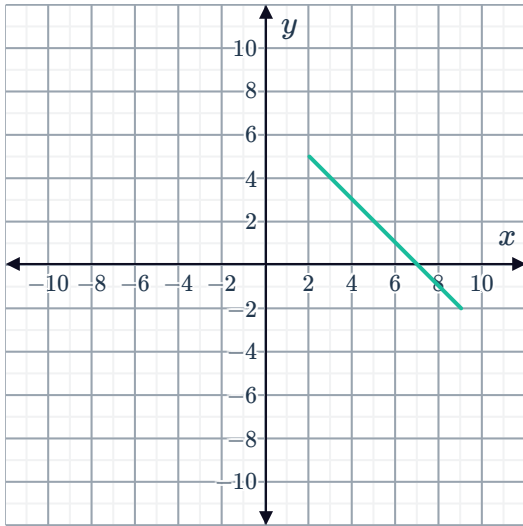
iv $[0, \infty)$

v



b

i



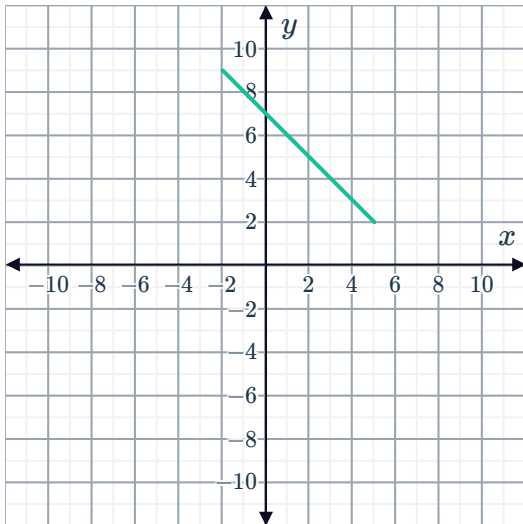
ii

$$f^{-1}(x) = 7 - x$$

iii $[-2, 5]$

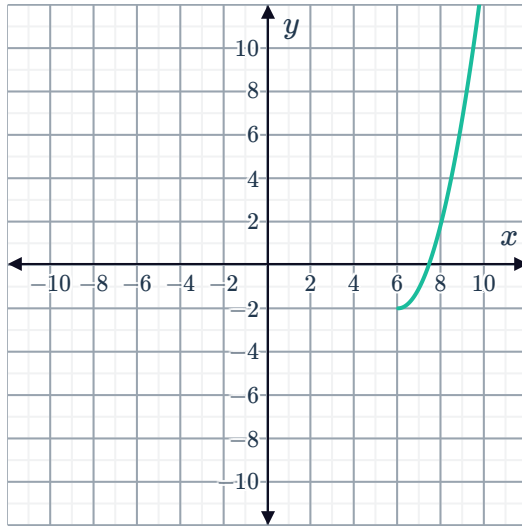
iv $[2, 9]$

v



c

i



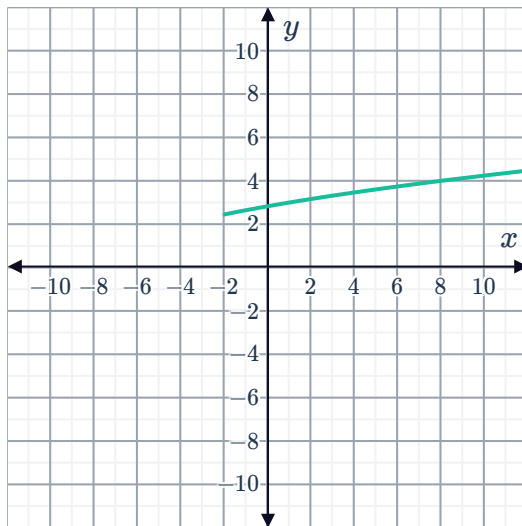
ii

$$f^{-1}(x) = \sqrt{x+2} + 6$$

iii $[-2, \infty)$

iv $[6, \infty)$

v



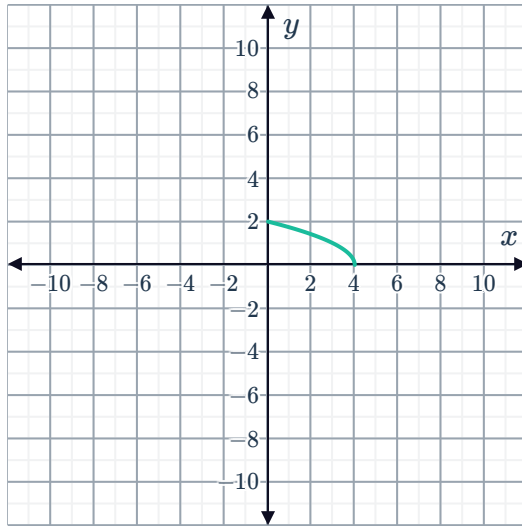
d

i



ii

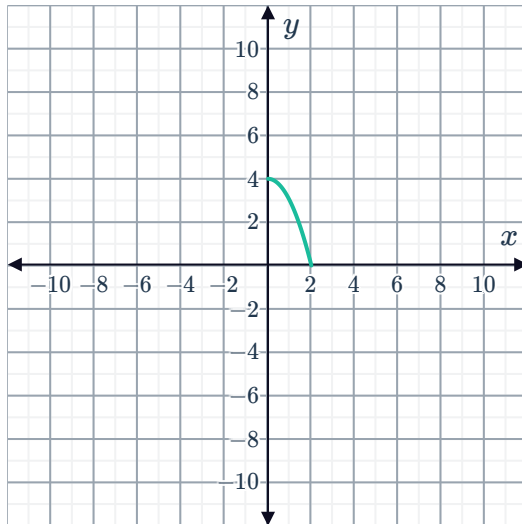
$$f^{-1}(x) = 4 - x^2$$



iii $(0, 2]$

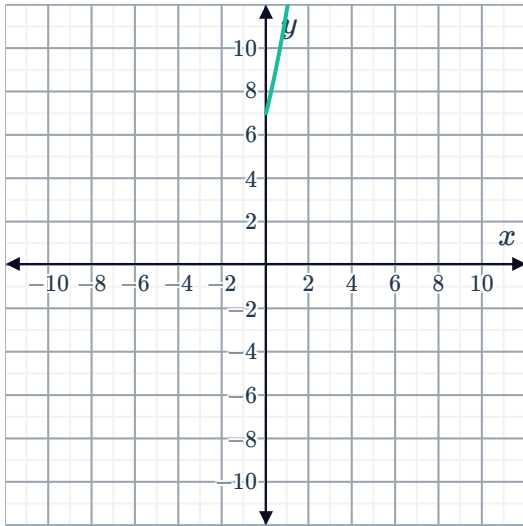
iv $[0, 4)$

v



e

i



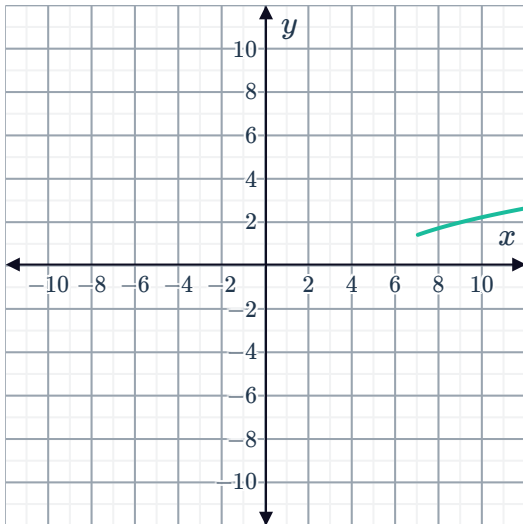
ii

$$f^{-1}(x) = \sqrt{x-3} - 2$$

iii $[7, \infty)$

iv $[0, \infty)$

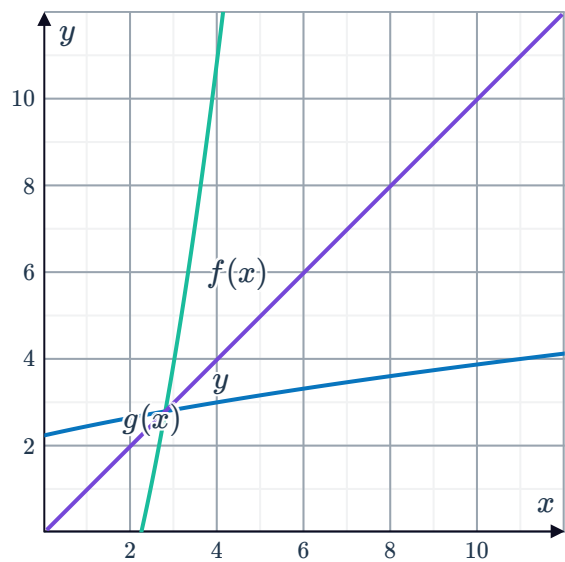
v



12

a $y = x$

b



c $f(x)$ is a reflection of $g(x)$ about $y = x$.



Applications

13

a $d = 4.9t^2$

b $d = 122.5$

14

a -25°

b 30°

c $T^{-1}(x) = \frac{9}{5}x + 32$

d For converting Celsius temperatures x to Fahrenheit temperatures $T^{-1}(x)$

15

a Yes

b $t(d) = \sqrt{\frac{120 - d}{4.9}}$

c No solution provided

d No solution provided

e Yes

f 4 seconds

16

a
$$f(x) = \begin{cases} (x-2)^2 - 6, & x < 2 \\ (x-2)^2 - 6, & x \geq 2 \end{cases}$$

b
$$g(x) = \begin{cases} g(x), & x \leq -1 \\ g(x), & -1 < x < 6 \\ g(x), & x \geq 6 \end{cases}$$